

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000824	Version:	4.0, CURRENT
	Status:	Effective	Effective Date:	12-Jul-2011
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DG-57 Service Manual

MOTORS REPLACEMENT

Procedure description
This procedure is a guideline to replace Centrifuge, Probe and Gripper Motors. <i>Note: Document REGD-0000824_v4 replaces document REGD-0000824_v3.</i>

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PROCEDURE 1: CENTRIFUGE MOTOR REPLACEMENT

EQUIPMENT

1.	232257 MOTOR CENTRIFUGE (w. Encoder, wire, plug & FW 2.0).
2.	Allen Keys, Philips screwdriver, Tube spanners, centrifuge rotor tube spanner (tools kit).

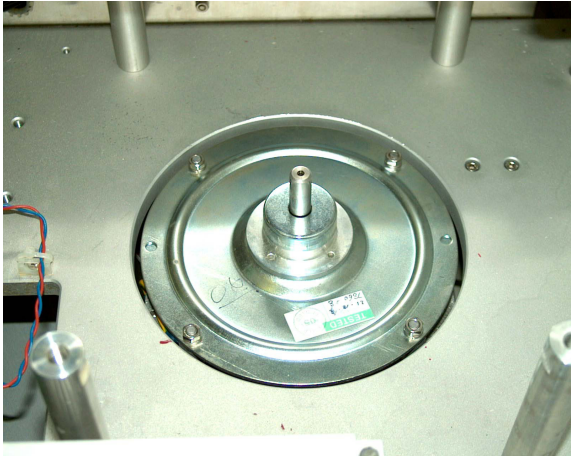
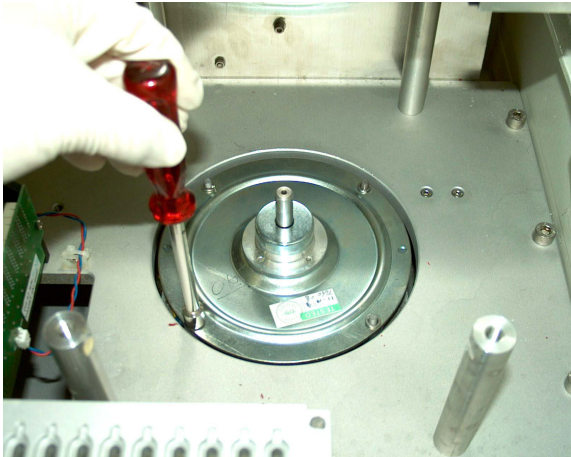
SET UP

1.	Turn the instrument off.
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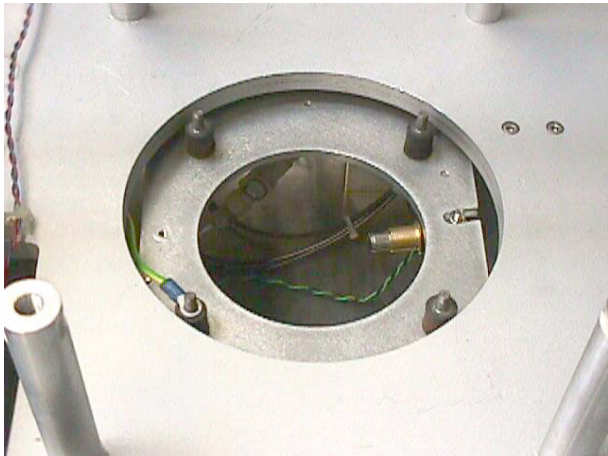
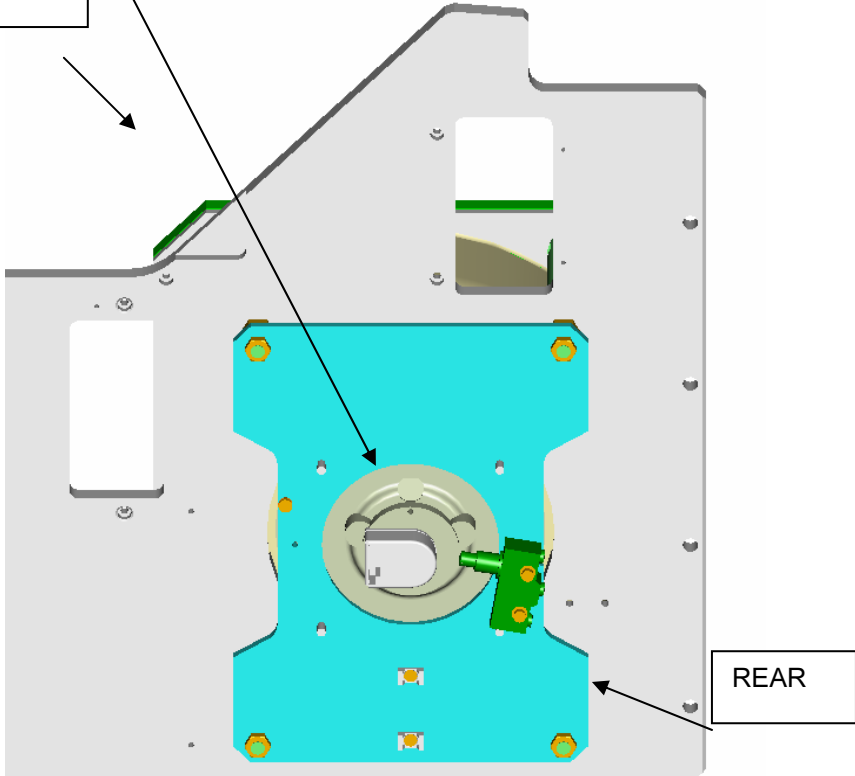
PROCEDURE

Process	Expected Results
1. The way a centrifuge motor is assembled may affect system performance. That procedure ensures that the motor sits flat on its basis and the silent block underneath are not twisted nor damaged. 2. Remove the top cover of the centrifuge. Picture 1 	--- ---
3. Remove all the card holders. 4. Remove the centrifuge rotor with the help of the special tube spanner (included in Wadiana Tools Kit). Picture 2 	--- ---

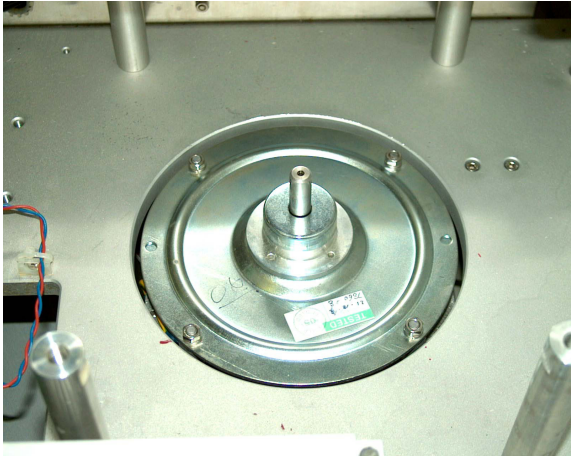
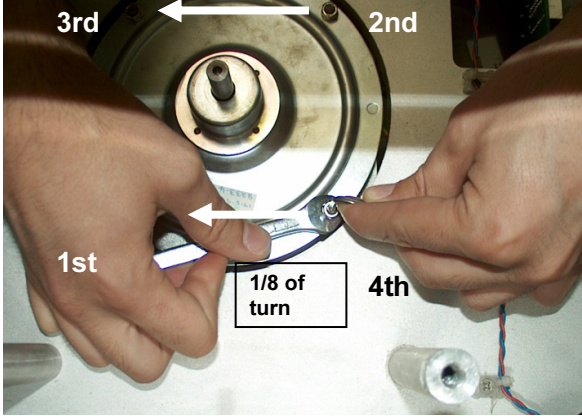
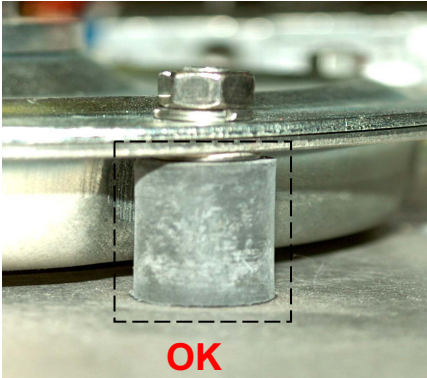
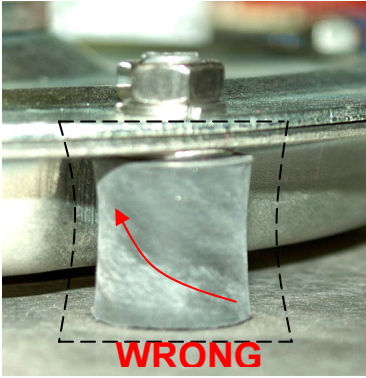
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Process	Expected Results
5. Remove the centrifuge pot to access the centrifuge motor. Picture 3 	---
6. Remove 4 nuts (& washers) with the help of a tube spanner. Picture 4 	---
7. Lift the motor as shown in the picture. Picture 5 	---

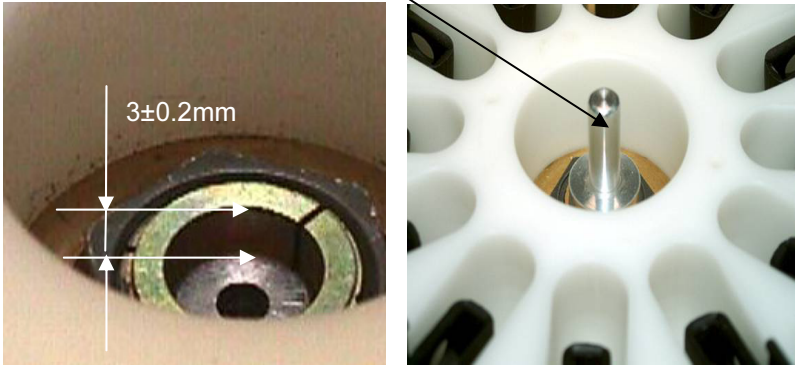
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Process	Expected Results
8. Unplug encoder connector. Make a mark on it to easy assembly afterwards. Unplug power connector. 9. Remove the motor.	Motor is ready to be removed
Picture 6 	---
10. Remove all silentblocks. Tight the new ones <u>manually</u>. 11. Plug encoder connector. Ensure it connected correctly. Plug power connector Place the new motor. <u>Note the correct position.</u>	---
Picture 7 FRONT  REAR	---

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Process	Expected Results
12. Place washers and nuts manually. 13. Tight the nuts with your fingers <u>as much as possible</u>. Picture 8  14. Screw the nuts in crossed way and very slowly (1/8 of turn every time (45°)) and repeat it until they are tighten, but not too much because you can twist silentblocks and then motor may be tilted, affecting unloading centrifuge function. Picture 9  15. Prevent the silentblock from rotating using an Allen key.  	--- --- --- ---

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Process	Expected Results
16. Reassemble centrifuge pot. 17. Insert centrifuge rotor as shown in the picture 10 and tight it with the help of the special tube spanner. Rotor must be assembled at 3.0 ± 0.2 mm by means of the <u>centrifuge height adjustment tool</u> Picture 10  Note: Apply REGD-0000801 procedure. 18. Insert again the card holders. 19. Place the top cover of the centrifuge. (see Picture 1) 20. After motor replacement, gripper and unbalance detector adjustment must be performed. Note: Apply REGD-0000803 procedure.	--- Rotor turns freely and it doesn't touch the bottom of the centrifuge pot. --- --- ---

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PROCEDURE 2: PROBE MOTOR REPLACEMENT

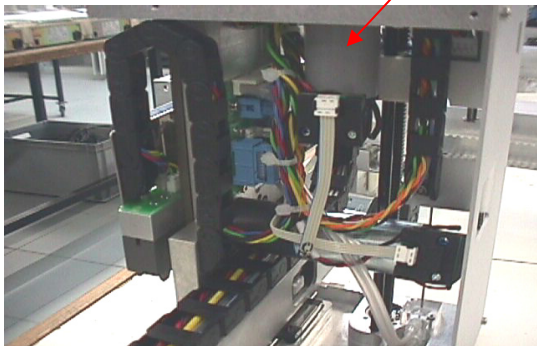
EQUIPMENT

3. 232089 MOTOR PROBE (with wire, plug and gear wheel)
4. Allen Keys, Philips screwdriver, Tube spanners.

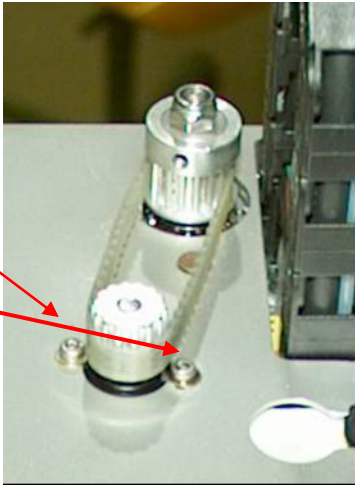
SET UP

2. Turn the instrument off.

PROCEDURE

Process	Expected Results
21. Remove covers. Remove the black top cover to access the belt easily. There are two screws subjecting it. Remove the back cover unscrewing the 4 screws. 22. Unplug the motor. Unplug encoder and power connectors. Note: pay attention to motor position (<u>encoder facing backwards</u>). If mounted otherwise it may interfere with moving parts. Picture 11 	--- ---
23. Remove the <u>two screws</u>. Picture 12 	The belt is free and the motor can be removed.
24. Remove the motor. 25. Insert the new motor. Put the belt in the pinion. 26. Insert the two screws but don't tighten them.	--- --- ---

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Process	Expected Results
27. Place the belt pushing the motor pinion while <u>screwing this screw first</u> Picture 13 Continue pushing the motor pinion while tightening the <u>screw in the slotted hole</u> Note: Take care not to overtorque the screws  ---	
28. Assemble the back cover	---
29. Assemble the black top cover:	---
30. Use the document REGD-0000805 to check the replacement.	---

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PROCEDURE 3: GRIPPER MOTOR REPLACEMENT

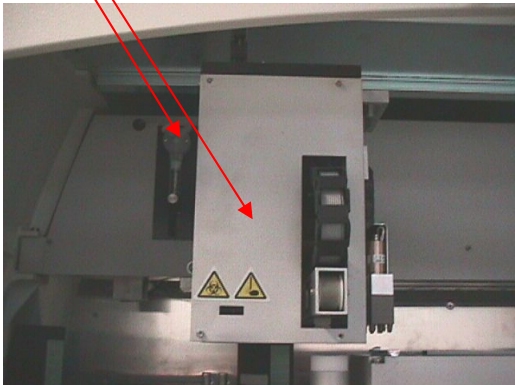
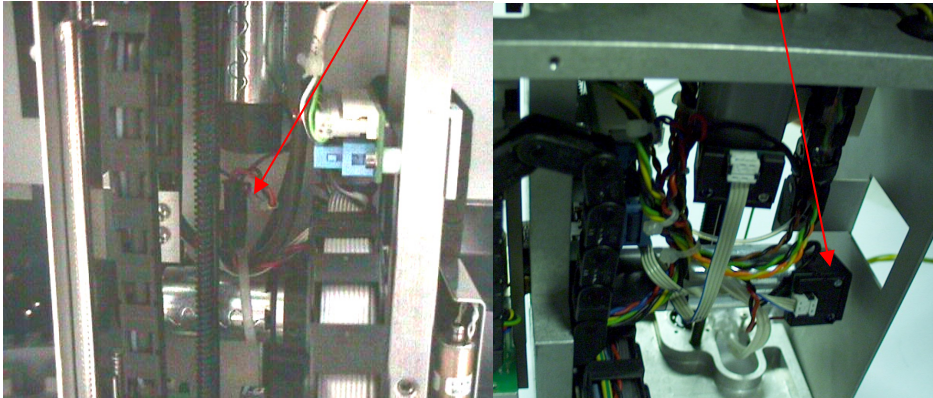
EQUIPMENT

1. 232090 MOTOR GRIPPER (with wire, plug and gear wheel)
2. Allen Keys, Philips screwdriver, Tube spanners.

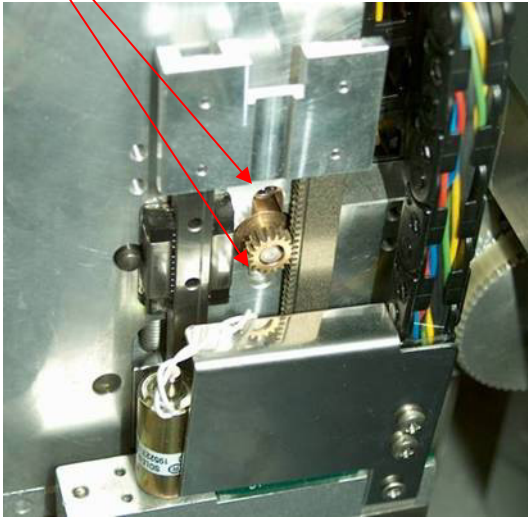
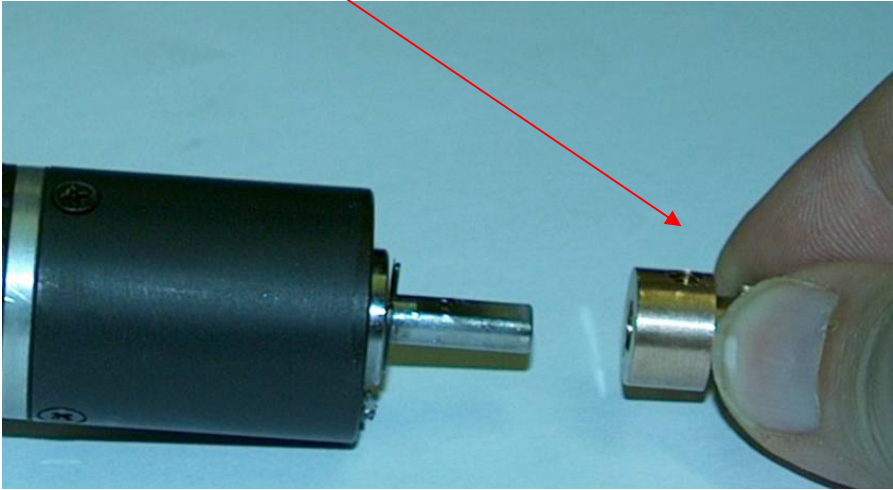
SET UP

1. Turn the instrument off.

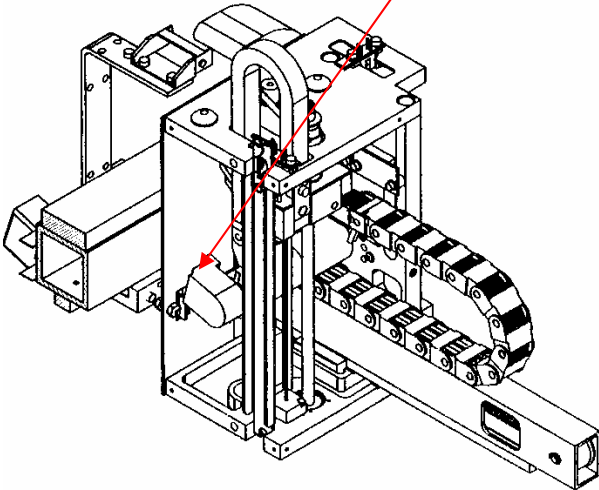
PROCEDURE

Process	Expected Results
1. Remove <u>front and left covers</u> unscrewing the 4 screws. Take care with: - The card treader (raise the probe). - The chain (is fixed in the left cover). - The spring. Picture 14 	---
2. Move down the Probe. Unplug the <u>power cable</u> and <u>encoder cable</u> of the motor. Picture 15 	---
3. Move the gripper to a position when the two screws that subject the motor are accessible.	The motor is ready to be removed

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Process	Expected Results
4. Remove the <u>two screws</u> Picture 16  5. Remove the motor 6. Remove the pinion from the old motor and insert it to the new one (if it is in bad condition put a new one). Unscrew the <u>set screw</u> and remove the pinion Picture 17  7. Insert the pinion so that the set screw concords with the flat surface of the motor axis.	--- --- --- ---

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Process	Expected Results
8. Insert the motor and pay attention to motor position (<u>encoder facing backwards</u>).It mounted otherwise it may interfere with moving parts. Fix the motor with the two screws. Picture 18 	---
9. Assemble the left cover: Raise the probe and assemble the card treader. Assemble the spring between the card treader and the left cover. Fix the cover with the four screws.	---
10. Assemble the frontal cover. Fix the cover with the four screws.	---
11. Use the document REGD-0000805 to check the replacement.	---

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DOCUMENT CHANGE HISTORY

Document change history			
Version	Date	Change	Rationale
2.0	2005/05/09	Document REGD-0000824_v2 replaces document 57051V0M000.	Document reference changed
3.0	2011/04/28	Added 'previous version 'data', 'change' and 'rationale' inside document change history section.	Update change history for previous document history modification.
3.0	2011/04/28	Updated the Document Change History: Date format changed to YYYY/MM/DD Dates changed to document version effective date.	Adoption of ISO 8601 date format to avoid ambiguity.
3.0	2011/04/28	ID-GELSTATION and PROVUE erased from the title.	Service manual intended for Grifols market.
3.0	2011/04/28	Procedure description : Document Document REGD-0000824_v3 replaces document REGD-0000824_v2.	Document version changed
3.0	2011/04/28	Modified spare part names and spare part codes inside 'equipment' sections.	Document change to update part numbers.
4.0	See document header.	<i>Document REGD-0000824_v4 replaces document REGD-0000824_v3.</i>	Document version changed.
4.0	See document header.	Document title.	Commercial reference substituted by project reference.
4.0	See document header.	Document header.	Increase information to easy document search.